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Short note

Spectroscopic analysis of Nd:BGSO crystal grown by the micro-pulling-down method

Na Li [a], Jianye Wang [b], Yanyan Xue [c], Qingsong Song [c], Xiaodong Xu [d,*], Dongzhen Li [d], Jun Xu [c,*]

[a] School of Mathematics and Physics, Jinzhong University, Jinzhong 030619, China
[b] Shanxi Yellow River Frontier Materials Research Institute, Taiyuan 030000, China
[c] School of Physics Science and Engineering, Institute for Advanced Study, Tongji University, Shanghai 200092, China
[d] Jiangsu Key Laboratory of Advanced Laser Materials and Devices, School of Physics and Electronic Engineering, Jiangsu Normal University, Xuzhou 221116, China

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ABSTRACT

A Nd:BGSO crystal with the dimensions of Φ2 × 131 mm^3 has been grown successfully by the micro-pulling-down method. The peak absorption cross-section at 808 nm was calculated to 0.98 × 10^{-20} cm^2 with full-width at half-maximum of 17.4 nm. The stimulated emission cross-section was obtained to be 3.15 × 10^{-20} cm^2 at 1063 nm. The fluorescence and radiative lifetimes of $^4F_{3/2}$ level were 248 μs and 300 μs respectively. All results indicate that the Nd: BGSO single crystal would be a promising near-IR solid-state laser material.

1. Introduction

In recent years, laser diode (LD) pumped solid-state lasers have been widely researched and developed due to their wide applications in medical treatment, nonlinear optics and scientific research. Most importantly, neodymium ions doped single crystals have attracted much attention owing to their intense absorption peaks at around 808 nm, and it is extremely appropriate for commercial laser diode GaAlAs pumping [1–6]. At the same time, Nd^{3+} is one of the most important ions in near-IR solid-state laser systems because the transition $^4F_{3/2}\rightarrow{}^4I_{11/2}$ (1064 nm) owns a big emission cross-section and a broad full-width at half-maximum (FWHM) [7–9].

Bismuth germanate ($Bi_4Ge_3O_{12}$, BGO) single crystal is a potential host medium in the area of solid-state lasers thanks to its magnificent thermo-mechanical properties. For instance, it owns a small thermal coefficient of expansion (6.3 × 10^{-6}/℃) and a reasonable thermal conductivity (2.59 W m^{-1} K^{-1}) [10,11]. Also, as a potential laser host crystal, Lin et al. demonstrated Nd^{3+}-doped BGO single-crystal fiber with an output power output power of 3.37 W and slope efficiency of 31.2% [12]. If doping a small amount of BSO into BGO, it does not change the structure of BGO crystals [13–16]. As a matter of fact, $Bi_4(Ge_xSi_{1-x})_3O_{12}$ (BGSO) single crystals have been grown by the Czochralski method [17], the vertical Bridgman method [14] and the micro-pulling-down method (μ-PD) [15]. Among them, the μ-PD technology has a great deal of characteristics, such as less raw materials, faster growth speed and adjustable diameter from tens of micron to millimeter. However, researches were only concentrated on the scintillating properties of the BGSO crystals, such as absolute light yield, the energy resolution and fluorescence decay time [18]. It is worth mentioning that

* Corresponding authors.
E-mail addresses: xdxu79@jsnu.edu.cn (X. Xu), xujun@mail.shcnc.ac.cn (J. Xu).

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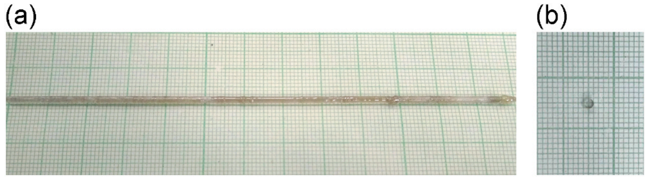

Fig. 1. (a) The as-grown Nd:BGSO crystal; (b) the cutting sample.

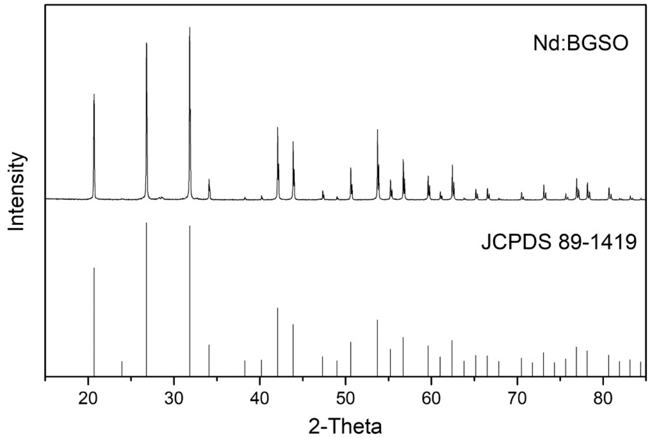

Fig. 2. XRD pattern of as-grown Nd:BGSO crystal.

until now only one paper reported the spectra characteristic of Dy^{3+} ions doped BGSO crystal [19]. Feng X. et al. reported the laser operation of Nd:$BG_{1-x}S_xO$ (Nd:BGSO) crystal was grown using the micro-pulling-down method for the first time, and the maximum output power was 1.038 W, and the slope efficiency was 31.3% [20]. And no investigations about the spectroscopic analysis of Nd: BGSO crystal have been yet investigated.

In this work, crystal growth, optical properties and JO theory analysis of Nd:BGSO single crystal were discussed in detail for the first time.

2. Experimental section

2.1. Crystal growth

The Nd:BGSO single crystal was successfully grown by the μ-PD technology in the air atmosphere. The Nd_2O_3, Bi_2O_3, GeO_2 and SiO_2 powders with 5 N purity were served as the raw materials and weighed accurately according to the chemical reaction $(Nd_{0.003}Bi_{0.997})_4(Ge_{0.95}Si_{0.05})_3O_{12}$. The powders were thoroughly mixed in an agate jar, sintered at 800 °C for 12 h and put in the platinum crucible. After that, the melt was pulled down along a micro via at the bottom of the platinum crucible with the pulling rate of 0.5 mm/min. The crystal of BGO with <211> orientation was used as seed. As shown in Fig. 1(a), the crystal with dimensions of $\Phi 2 \times 131$ mm^3 was obtained and the dark brown color on the surface of the crystal was due to volatilization of Bi_2O_3 during the crystal growth, which is accordance with that of BGSO grown by the vertical Bridgman method [16]. The polished sample was highly transparent and free from cracks, which is showed in Fig. 1(b).

2.2. Spectra measurements

It is essential to verify the pattern of as-grown Nd:BGSO crystal. We cut a small piece of the sample from the crystal and then ground up into a fine powder in a quartz mortar. After that the structure of Nd:BGSO crystal was determined by X-ray powder diffraction (XRD), and the scan width was 0.02° within the range of 15°–85°. The micro-photoluminescence (μ-PL) spectra was taken by the NIR spectrometer, and which was equipped with InGaAs CCD imaging sensor. The absorption spectrum of Nd:BSGO crystal was recorded using a Lambda 900 spectrophotometer within the range of 340–900 nm and the step speed was set as 1.0 nm. The emission spectrum and the decay time of Nd:BSGO crystal were performed by a FSP920 spectrophotometer under 808 nm LD exciting.

3. Results and discussion

The structure of the Nd:BGSO crystal is presented in Fig. 2 and all diffraction peaks are in good agreement with the standard Joint

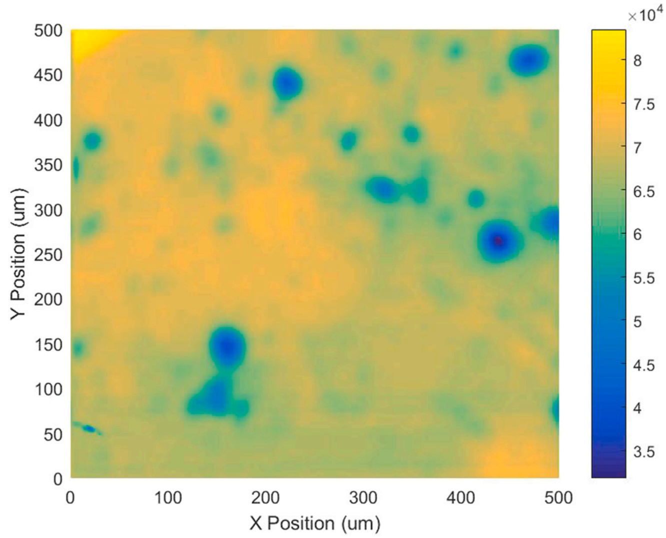

Fig. 3. The micro-photoluminescence map of Nd:BGSO crystal.

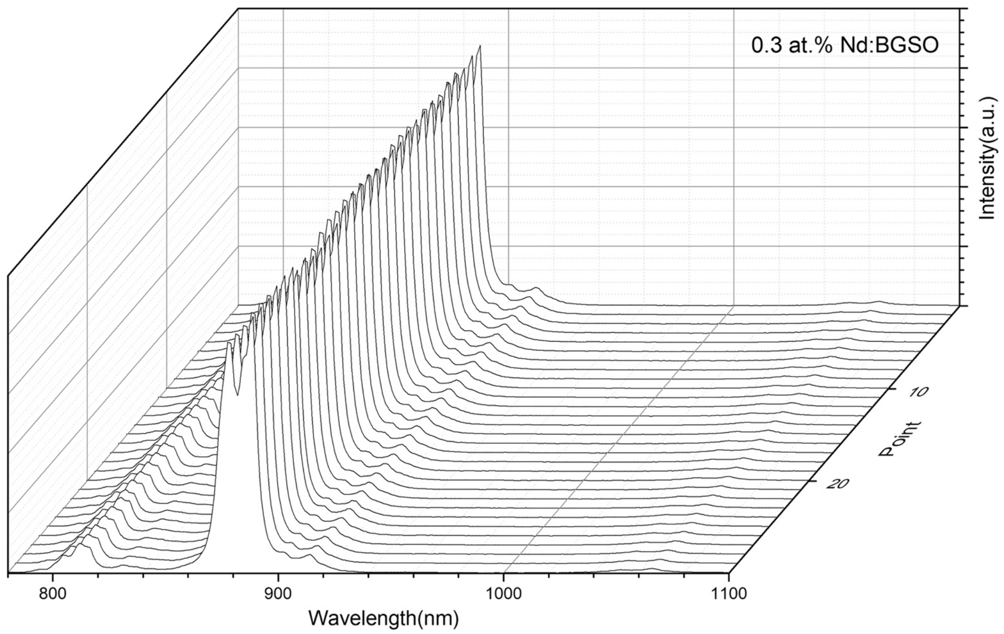

Fig. 4. The micro-photoluminescence spectra of Nd:BGSO crystal along the diameter.

Committee on Powder Diffraction Standards (JCPDS) card of $Bi_4Ge_3O_{12}$ (PDF#89-1419). The data confirm that the structure of Nd: BGSO is the isometric system and the space group is I43d. According to above result, the lattice constants can be estimated using the least square method. And the value was calculated to be 10.513 Å, which was larger than that of Nd doped BGO crystal (10.503 Å) [21].

The Nd^{3+} ions distribution in the BGSO crystal were assessed by the μ-PL intensity on the entire wafer. The 532 nm laser was focused on the crystal surface through the objective lens, and the diameter of the spot is about 500 nm. The fluorescence spectra of each spot were measured by moving the spot position along the diameter. The numbers in the experimental data were recorded from left to right at roughly the same distance. As shown in Fig. 3, there are a few regions with low fluorescence intensity, that may be due to the small holes in the crystal surface, and a part of laser is scattered back into the air by the hole. The fluorescence spectra along the diameter are presented in Fig. 4. There is very slight difference about emission intensity of 30 points. The μ-PL imaging illustrates that the distribution of the neodymium ions in the BGSO crystal is well-proportioned, which is consistent with the report in the Nd:BGO single crystal [22].

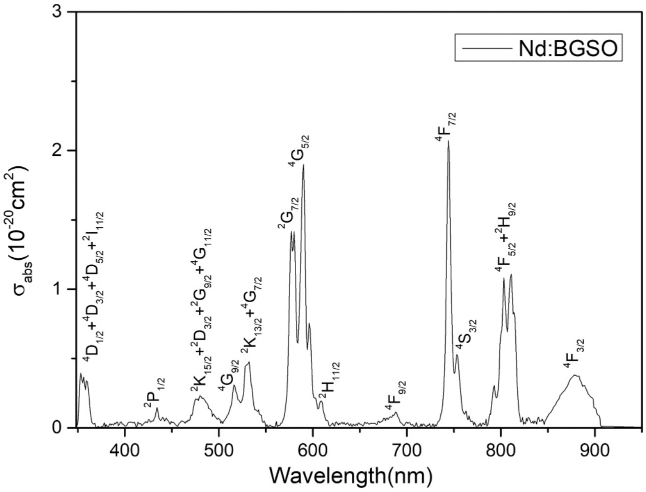

Fig. 5. The absorption spectrum of Nd:BGSO crystal.

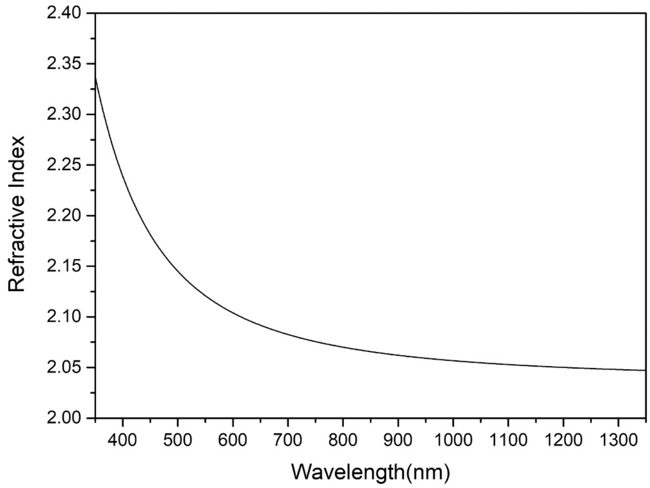

Fig. 6. The refractive index of Nd:BGSO crystal.

The room temperature absorption spectrum of Nd:BGSO crystal in the range of 340–950 nm is presented in Fig. 5, and absorption bands connected with the transitions from the ground state $^4I_{9/2}$ to the excited states of Nd ions are marked. The absorption band at around 808 nm for $^4I_{9/2}\rightarrow{}^4F_{5/2}+{}^2H_{9/2}$ transition is important for the Nd ions doped crystals, which can be pumped by the available high-power AlGaAs laser diodes. The absorption cross-section at 808 nm was calculated to be 0.98×10^{-20} cm^2 and the FWHM is about 17.4 nm, which is broader than Nd:BGO crystal (16.0 nm) [21]. Therefore, the Nd:BGSO crystal is not limited to the temperature stability of the output wavelength of the diode laser, and it is not difficult for Nd:BGSO crystal to absorb adequate energy from the AlGaAs pumping source.

So far, the Judd–Ofelt (JO) theory is a commonly practical method to calculate the experimental oscillator strength and theoretical oscillator strength of electric dipole (ED) transitions between multiplet component for various rare earth ions in host materials, including glasses and crystals [23,24]. Specifically, the experimental oscillator strength can be reached from the absorption transitions by a fiendishly clever equation:

$$S_{\exp}\left(J \rightarrow J'\right) = \frac{3hc(2J+1)}{8\pi^3 e^2 N} \frac{9n}{\left(n^2+2\right)^2} \frac{\ln 10}{\overline{\lambda} L} \int OD(\lambda) d(\lambda) \tag{1}$$

where $\overline{\lambda}$ indicates the mean wavelength of the selected absorption band, h is Planck's constant, c represents the velocity of light, e is the electronic charge, L is the thickness of double-sided polished sample. And the refractive index (n) of the BGSO crystal was measured by elliptical polarization spectrometer in the wavelength range from 350 to 1350 nm and depicted in Fig. 6. The three J-O intensity parameters Ω_t(t = 2,4,6) can be achieved through least square fitting method.

The calculated line strength $S_{cal}(J \rightarrow J')$ can be calculated by the formula:

Table 1
Measured and calculated line strengths ($\times 10^{-20}$ cm^2) of Nd:BGSO crystal.

Transitions from $^4I_{9/2} \to$	$\bar{\lambda}$(nm)	n	$S_{exp}(10^{-20}cm^2)$	$S_{cal}(10^{-20}cm^2)$
$^4D_{1/2}+^4D_{3/2}+^4D_{5/2}+^2I_{11/2}$	360	2.313	0.598	0.846
$^2P_{1/2}$	433	2.197	0.050	0.053
$^2D_{3/2}+^2G_{9/2}+^4G_{11/2}+^2K_{15/2}$	480	2.158	0.366	0.144
$^4G_{9/2}$	515	2.136	0.077	0.148
$^2K_{13/2}+^4G_{7/2}$	532	2.129	0.247	0.445
$^2G_{7/2}$	577	2.111	0.868	0.364
$^4G_{5/2}$	591	2.107	0.985	1.041
$^2H_{11/2}$	610	2.101	0.031	0.028
$^4F_{9/2}$	683	2.085	0.075	0.102
$^4F_{7/2}$	744	2.076	0.791	1.041
$^4S_{3/2}$	754	2.075	0.195	0.548
$^4F_{5/2}+^2H_{9/2}$	806	2.069	1.876	1.540
$^4F_{3/2}$	877	2.064	0.341	0.447
RMSΔS		0.269		

Table 2
The Judd–Ofelt strength parameters of neodymium doped different crystals.

Crystals	$\Omega_2(10^{-20}$ cm$^2)$	$\Omega_4(10^{-20}$ cm$^2)$	$\Omega_6(10^{-20}$ cm$^2)$	Ω_4/Ω_6	Ref.
BGSO	0.43	1.40	2.31	0.61	This work
BGO	1.50	3.02	3.15	1.31	[21]
YAG	0.62	1.70	5.76	0.29	[26]
YAB	1.79	2.44	3.25	0.75	[27]
LGSO	5.37	1.63	5.57	0.29	[28]
$LuVO_4$	7.18	5.99	5.78	1.03	[29]
Lu_2SiO_5	2.59	4.90	5.96	0.82	[30]
$LiLuF_4$	1.38	1.96	5.05	0.39	[31]
$Ca_3Ga_2Ge_3O_{12}$	0.51	3.68	4.59	0.80	[32]

$$S_{\mathrm{cal}}\left(J \to J'\right) = \sum_{t=2,4,6} \Omega_t |\langle 4f^n[S,L]J||U^{(t)}||4f^n[S',L']J'\rangle|^2 \tag{2}$$

where $\langle||U^{(t)}||\rangle$ means the squared reduced matrix elements, which can be found from reference documentation [25]. It is well known that it is almost independent with the hosts.

The root-mean-square (RMS) deviation between the experimental oscillator strength and theoretical oscillator strength, is exactly expressed as:

$$rms\Delta s = \sqrt{\frac{\sum_{J'}\left(S_{\mathrm{exp}} - S_{cal}\right)^2}{N_0 - 3}} \tag{3}$$

where N_0 represents the amount of transitions.

As showed in Table 1, the value of the root-mean-square (RMS) deviation was calculated to be 0.269×10^{-20} cm^2, and a lower value of RMS confirms that the theoretical data are in acceptable agreement with the experimental results. The JO intensity parameters of Nd:BGSO and other Nd^{3+}-doped crystals are listed in Table 2. The results demonstrate that the value of Ω_2 is relatively small than that of other Nd^{3+}-doped crystals. So, the bond covalency of the Nd–O in BGSO single crystal is comparatively weaker. The spectroscopic quality factor Ω_4/Ω_6 for Nd:BGSO crystal is equal to be 0.61 and the value is similar to Nd: $Ca_3Ga_2Ge_3O_{12}$, Lu_2SiO_5, and $YAl_3(BO_3)_4$ crystals, and much larger than that of other neodymium ions doped materials, which implies that the Nd:BGSO crystal could be a very up-and-coming laser material.

In general, the obtained J–O intensity parameters were used to compute the radiation transition probability, fluorescence branching ratio and experimentally lifetime for the excited state according the fluorescent emission spectra. The radiative transition rates $A(J \to J')$ can be expressed as:

$$A = A_{ed} + A_{md} \tag{4}$$

$$A_{ed}\left(J \to J'\right) = \frac{64\pi^4 e^2}{3h(2J+1)\bar{\lambda}^3} \frac{n(n^2+2)^2}{9} S_{ed} \tag{5}$$

Table 3
Radiative transition rates, branching ratios and radiative lifetime for different transition levels of Nd:BGSO crystal.

Transition	$\overline{\lambda}$(nm)	$A_a(s^{-1})$	β_a(%)
$^4F_{3/2}\rightarrow{}^4I_{9/2}$	885	1195	35.96
$^4F_{3/2}\rightarrow{}^4I_{11/2}$	1063	1838	52.27
$^4F_{3/2}\rightarrow{}^4I_{13/2}$	1341	372	11.20
$^4F_{3/2}\rightarrow{}^4I_{15/2}$	1880	19	0.57
Radiative lifetime (ms)	0.300		

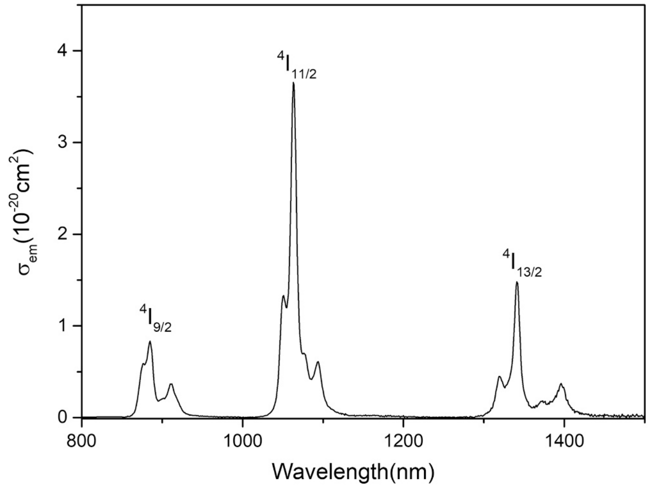

Fig. 7. Fluorescence spectrum of Nd:BGSO crystal.

$$A_{md}\left(J\rightarrow J^{'}\right)=\frac{64\pi^4e^2n^3}{3h(2J+1)\overline{\lambda}^3}S_{md} \tag{6}$$

Where the line strength S_{md} often is not classified as a variable, but a value, and S_{ed} can defined by:

$$S_{ed}\left(J\rightarrow J^{'}\right)=\sum\Omega_t|\langle J\|U^t\|J^{'}\rangle|^2, t=2,4,6 \tag{7}$$

where $\|U^t\|$ is the emission transition element [33].

The fluorescence branching ratios $\beta(J\rightarrow J^{'})$ and radiation lifetime $\tau_{\rm rad}$ can be indicated using the following equations:

$$\beta\left(J\rightarrow J^{'}\right)=\frac{A(J\rightarrow J^{'})}{\sum_{J^{'}}A(J\rightarrow J^{'})} \tag{8}$$

$$\tau_{\rm rad}=\frac{1}{\sum_{J^{'}}A(J,J^{'})} \tag{9}$$

The transition probability, fluorescence decay rates and the radiation lifetime of $^4F_{3/2}$ lever in Nd:BGSO single crystal were recorded in Table 3. The radiative lifetime of $^4F_{3/2}$ level was calculated to be 300 μs. In general, fluorescence decay rates can be useful to forecast the comparative strengths of whole emission lines from an upper state to its lower state. In the Nd:BGSO crystal, the transition of $^4F_{3/2}\rightarrow{}^4I_{11/2}$ owns the maximum branching ratio in relation to other transitions and the value is 52.27%, which suggests that the laser emission in 1064 nm has the highest possibility.

The emission spectrum of Nd:BGSO single crystal was recorded at room temperature. As showed in Fig. 7, there are three main emission peaks centered at 885 nm, 1063 nm and 1341 nm, which are corresponding to the transitions of $^4F_{3/2}\rightarrow{}^4I_{9/2}$, $^4F_{3/2}\rightarrow{}^4I_{11/2}$ and $^4F_{3/2}\rightarrow{}^4I_{13/2}$, respectively. And most important of all, the strongest peak is situated at 1063 nm, which is in accordance with the conclusions obtained in Table 3. The stimulated emission cross-section can be expressed by:

$$\sigma_{em}(\lambda)_{JJ^{'}}=\frac{\lambda^5}{8\pi n^2c\tau_{\rm rad}}\frac{I(\lambda)}{\int\lambda I(\lambda)d\lambda} \tag{10}$$

where λ is the emission wavelength, and $I(\lambda)$ represents the experimental fluorescent intensity. It is noteworthy that the emission cross-section is 3.15×10^{-20} cm^2 with the FWHM of 9.9 nm at 1063 nm, which is higher than that of 3.0×10^{-20} cm^2 reported for Nd:BGO

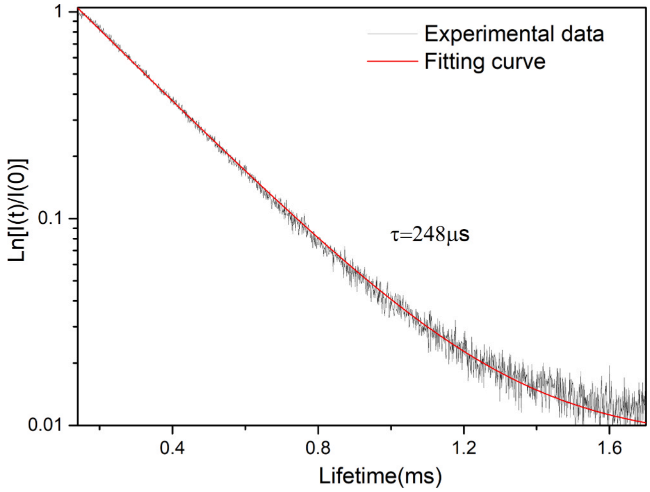

Fig. 8. Decay curve of the $^4F_{3/2}$ level of Nd:BGSO crystal.

crystal [7]. The result indicates that Nd:BGSO would be a hopeful crystalline solid for 1064 nm laser output.

The fluorescence decay curve of the $^4F_{3/2}$ level at the wavelength of 1063 nm under excitation at 808 nm is given in Fig. 8. The fluorescence lifetime was obtained to be 248 μs by a single exponential fitting. The value is somewhat higher than that of Nd:BGO crystal (220 μs) [7]. Furthermore, the quantum efficiency ($\eta = \tau_f/\tau_{rad}$) was estimated to be 82.7%, which is higher than that of Nd:YVO_4 crystal (72.0%) and Nd:$LuVO_4$ crystal (74.0%) [29]. The result shows that Nd:BGSO single crystal can be a auspicious medium for laser power output.

4. Conclusion

The Nd:BGSO single crystal was successfully grown by the micro-pulling-down method. The peak absorption cross section at 808 nm is 0.98×10^{-20} cm^2 with a FWHM of 17.4 nm, which is appropriate for AlGaAs laser diode pumping. The Judd–Ofelt theory was used to forecasting its potential of laser output. The intensity parameters $\Omega_{2,4,6}$ were calculated to be 0.43×10^{-20} cm^2, 1.40×10^{-20} cm^2 and 2.31×10^{-20} cm^2, respectively. The largest stimulated emission cross-section of 1063 nm is 3.15×10^{-20} cm^2. The lifetime of the $^4F_{3/2}$ level was obtained to be 248 μs. All the results state clearly that Nd:BGSO single crystal could be a potential material for near-IR laser output.

Declaration of Competing Interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled.

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